

- Check if $\log Z$ Sol
is correctly assigned
(right now it's Fe
only ;

- $\log (M/M_{\odot}) = 11$ very massive quiescent
galaxies ; but extremely metal poor
(we expect at least $\log Z = 0$
roughly solar

- Think there's an offset ; redshift wrong ;
absorption lines don't seem to match up)

- Magnesium B poorly fit

- Polynomial order :

We want { Rule of thumb $\frac{\text{Wavelength range}}{100}$ (empirical; no physics)
models not more than $100 \text{ \AA}^0$

- Start with 5, 10, etc for polynomial
- Do a sanity check on polynomial to see if it curves

- If polynomial order is too high it mimics spectral signals; if too low flux calibration

Local elliptical galaxies

$$\alpha / \text{Fe} \approx 0.4$$

$$\log z_{\text{solar}} \approx 0.00$$

We expect Legn C to be similar to current elliptical galaxies

(no significant chemical evolution)

↳ so this is surprising

• Mg-B is not so well fitted (?)

Look into it

• Systematic uncertainties in photometry (is it present?)
Add calibration uncertainties

• Ceridwen changed metallicity definition?

• $\log Z$ is absolute not solar!?
so actually the error is
not so bad

• Conversion $+8.95$?

• But α/Fe should still be $+0.4$?

• She will change coordinates to solar metallicity

• $\alpha\text{-Mc}$ $\log Z$ sol is only
iron to hydrogen

↳ Look into that

• solar metallicity is only

-0.8

• Fe/H (90% hits upper band)

↓
log is too high
for other model

↓
It's hitting upper bound
of prior

① Look into polynomial

② Look into metallicity
to see if it matches
↳ And fix metallicity
definition

③ noise calibration

④ plot Mass-metallicity

⑤ Slack channel

↳ post in a channel in the group

⑥ In principle more accurate age measurements would be super good

⑦ Grab one beautiful case

↳ High S/N; strong absorption features

↳ This is a good thing to compare against literature

⑧ You should fit for velocity dispersion

⑧ Once confident you are

↓
as a broadening parameter

↓
even though

lega - C is
already accurate

④ Make slack channel with

① Amanda, MJ, Sandro

② List all free parameters
and priors

⑤ What to check against
in literature?

① Check velocity dispersion

② Ages, published metallicity

↙ ③ Check against the Terry

This is
modelled in

literature already

(Legu - C papers)

prospector fnt

④ Star formation history
prior?



Use a lot of
methods; find
which are rigorous